

Primary 4-6

INGREDIENT LAB ACTIVITY SYSTEM

IL-201 to IL-300

QUESTION -> SAMPLE -> MEASURE -> ANALYZE -> EXPLAIN

Purpose	Build food and ingredient literacy by connecting plant structure, observation, measurement, post-harvest change, quality and product-development thinking.
Ingredient set	Hibiscus, ginger, turmeric, pineapple, citrus, mint, moringa, watermelon, tomato and maize.
Core rule	The Ingredient Lab is an educational investigation environment. It does not require tasting, medical claims or uncontrolled food processing.
Connections	Link each activity to GrowMeal vocabulary, journals, practical cards, Farm-to-Food activities, missions and assessment.

INGREDIENT LAB - HIBISCUS | PLANT PART: CALYX

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-201 - HIBISCUS - IDENTITY & PLANT PART

Plant part	Calyx
Learning objective	Investigate identity & plant part in hibiscus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-202 - HIBISCUS - EXTERNAL STRUCTURE

Plant part	Calyx
Learning objective	Investigate external structure in hibiscus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-203 - HIBISCUS - INTERNAL STRUCTURE

Plant part	Calyx
Learning objective	Investigate internal structure in hibiscus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-204 - HIBISCUS - COLOUR & TEXTURE

Plant part	Calyx
Learning objective	Investigate colour & texture in hibiscus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-205 - HIBISCUS - MASS & SIZE

Plant part	Calyx
Learning objective	Investigate mass & size in hibiscus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-206 - HIBISCUS - WATER/CHANGE OBSERVATION

Plant part	Calyx
Learning objective	Investigate water/change observation in hibiscus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-207 - HIBISCUS - SORTING & QUALITY

Plant part	Calyx
Learning objective	Investigate sorting & quality in hibiscus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-208 - HIBISCUS - POST-HARVEST CHANGE

Plant part	Calyx
Learning objective	Investigate post-harvest change in hibiscus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-209 - HIBISCUS - INGREDIENT-TO-FOOD PATHWAY

Plant part	Calyx
Learning objective	Investigate ingredient-to-food pathway in hibiscus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-210 - HIBISCUS - INVESTIGATION CHALLENGE

Plant part	Calyx
Learning objective	Investigate investigation challenge in hibiscus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - GINGER | PLANT PART: RHIZOME

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-211 - GINGER - IDENTITY & PLANT PART

Plant part	Rhizome
Learning objective	Investigate identity & plant part in ginger using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-212 - GINGER - EXTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Investigate external structure in ginger using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
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AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-213 - GINGER - INTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Investigate internal structure in ginger using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
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Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-214 - GINGER - COLOUR & TEXTURE

Plant part	Rhizome
Learning objective	Investigate colour & texture in ginger using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
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Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-215 - GINGER - MASS & SIZE

Plant part	Rhizome
Learning objective	Investigate mass & size in ginger using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
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Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-216 - GINGER - WATER/CHANGE OBSERVATION

Plant part	Rhizome
Learning objective	Investigate water/change observation in ginger using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
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AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-217 - GINGER - SORTING & QUALITY

Plant part	Rhizome
Learning objective	Investigate sorting & quality in ginger using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
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Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-218 - GINGER - POST-HARVEST CHANGE

Plant part	Rhizome
Learning objective	Investigate post-harvest change in ginger using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
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Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-219 - GINGER - INGREDIENT-TO-FOOD PATHWAY

Plant part	Rhizome
Learning objective	Investigate ingredient-to-food pathway in ginger using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
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Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-220 - GINGER - INVESTIGATION CHALLENGE

Plant part	Rhizome
Learning objective	Investigate investigation challenge in ginger using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
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AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - TURMERIC | PLANT PART: RHIZOME

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-221 - TURMERIC - IDENTITY & PLANT PART

Plant part	Rhizome
Learning objective	Investigate identity & plant part in turmeric using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-222 - TURMERIC - EXTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Investigate external structure in turmeric using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
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AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-223 - TURMERIC - INTERNAL STRUCTURE

Plant part	Rhizome
Learning objective	Investigate internal structure in turmeric using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
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Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-224 - TURMERIC - COLOUR & TEXTURE

Plant part	Rhizome
Learning objective	Investigate colour & texture in turmeric using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-225 - TURMERIC - MASS & SIZE

Plant part	Rhizome
Learning objective	Investigate mass & size in turmeric using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-226 - TURMERIC - WATER/CHANGE OBSERVATION

Plant part	Rhizome
Learning objective	Investigate water/change observation in turmeric using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-227 - TURMERIC - SORTING & QUALITY

Plant part	Rhizome
Learning objective	Investigate sorting & quality in turmeric using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-228 - TURMERIC - POST-HARVEST CHANGE

Plant part	Rhizome
Learning objective	Investigate post-harvest change in turmeric using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-229 - TURMERIC - INGREDIENT-TO-FOOD PATHWAY

Plant part	Rhizome
Learning objective	Investigate ingredient-to-food pathway in turmeric using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-230 - TURMERIC - INVESTIGATION CHALLENGE

Plant part	Rhizome
Learning objective	Investigate investigation challenge in turmeric using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - PINEAPPLE | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-231 - PINEAPPLE - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Investigate identity & plant part in pineapple using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-232 - PINEAPPLE - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Investigate external structure in pineapple using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-233 - PINEAPPLE - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Investigate internal structure in pineapple using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-234 - PINEAPPLE - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Investigate colour & texture in pineapple using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-235 - PINEAPPLE - MASS & SIZE

Plant part	Fruit
Learning objective	Investigate mass & size in pineapple using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-236 - PINEAPPLE - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Investigate water/change observation in pineapple using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-237 - PINEAPPLE - SORTING & QUALITY

Plant part	Fruit
Learning objective	Investigate sorting & quality in pineapple using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-238 - PINEAPPLE - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Investigate post-harvest change in pineapple using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-239 - PINEAPPLE - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Investigate ingredient-to-food pathway in pineapple using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-240 - PINEAPPLE - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Investigate investigation challenge in pineapple using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - CITRUS | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-241 - CITRUS - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Investigate identity & plant part in citrus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-242 - CITRUS - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Investigate external structure in citrus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-243 - CITRUS - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Investigate internal structure in citrus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-244 - CITRUS - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Investigate colour & texture in citrus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-245 - CITRUS - MASS & SIZE

Plant part	Fruit
Learning objective	Investigate mass & size in citrus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-246 - CITRUS - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Investigate water/change observation in citrus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-247 - CITRUS - SORTING & QUALITY

Plant part	Fruit
Learning objective	Investigate sorting & quality in citrus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-248 - CITRUS - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Investigate post-harvest change in citrus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-249 - CITRUS - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Investigate ingredient-to-food pathway in citrus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-250 - CITRUS - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Investigate investigation challenge in citrus using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - MINT | PLANT PART: LEAF

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-251 - MINT - IDENTITY & PLANT PART

Plant part	Leaf
Learning objective	Investigate identity & plant part in mint using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-252 - MINT - EXTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Investigate external structure in mint using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-253 - MINT - INTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Investigate internal structure in mint using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-254 - MINT - COLOUR & TEXTURE

Plant part	Leaf
Learning objective	Investigate colour & texture in mint using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-255 - MINT - MASS & SIZE

Plant part	Leaf
Learning objective	Investigate mass & size in mint using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-256 - MINT - WATER/CHANGE OBSERVATION

Plant part	Leaf
Learning objective	Investigate water/change observation in mint using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-257 - MINT - SORTING & QUALITY

Plant part	Leaf
Learning objective	Investigate sorting & quality in mint using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-258 - MINT - POST-HARVEST CHANGE

Plant part	Leaf
Learning objective	Investigate post-harvest change in mint using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-259 - MINT - INGREDIENT-TO-FOOD PATHWAY

Plant part	Leaf
Learning objective	Investigate ingredient-to-food pathway in mint using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-260 - MINT - INVESTIGATION CHALLENGE

Plant part	Leaf
Learning objective	Investigate investigation challenge in mint using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - MORINGA | PLANT PART: LEAF

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-261 - MORINGA - IDENTITY & PLANT PART

Plant part	Leaf
Learning objective	Investigate identity & plant part in moringa using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-262 - MORINGA - EXTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Investigate external structure in moringa using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-263 - MORINGA - INTERNAL STRUCTURE

Plant part	Leaf
Learning objective	Investigate internal structure in moringa using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-264 - MORINGA - COLOUR & TEXTURE

Plant part	Leaf
Learning objective	Investigate colour & texture in moringa using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-265 - MORINGA - MASS & SIZE

Plant part	Leaf
Learning objective	Investigate mass & size in moringa using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-266 - MORINGA - WATER/CHANGE OBSERVATION

Plant part	Leaf
Learning objective	Investigate water/change observation in moringa using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-267 - MORINGA - SORTING & QUALITY

Plant part	Leaf
Learning objective	Investigate sorting & quality in moringa using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-268 - MORINGA - POST-HARVEST CHANGE

Plant part	Leaf
Learning objective	Investigate post-harvest change in moringa using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-269 - MORINGA - INGREDIENT-TO-FOOD PATHWAY

Plant part	Leaf
Learning objective	Investigate ingredient-to-food pathway in moringa using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-270 - MORINGA - INVESTIGATION CHALLENGE

Plant part	Leaf
Learning objective	Investigate investigation challenge in moringa using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - WATERMELON | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-271 - WATERMELON - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Investigate identity & plant part in watermelon using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-272 - WATERMELON - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Investigate external structure in watermelon using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-273 - WATERMELON - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Investigate internal structure in watermelon using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-274 - WATERMELON - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Investigate colour & texture in watermelon using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-275 - WATERMELON - MASS & SIZE

Plant part	Fruit
Learning objective	Investigate mass & size in watermelon using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-276 - WATERMELON - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Investigate water/change observation in watermelon using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-277 - WATERMELON - SORTING & QUALITY

Plant part	Fruit
Learning objective	Investigate sorting & quality in watermelon using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-278 - WATERMELON - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Investigate post-harvest change in watermelon using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-279 - WATERMELON - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Investigate ingredient-to-food pathway in watermelon using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-280 - WATERMELON - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Investigate investigation challenge in watermelon using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - TOMATO | PLANT PART: FRUIT

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-281 - TOMATO - IDENTITY & PLANT PART

Plant part	Fruit
Learning objective	Investigate identity & plant part in tomato using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-282 - TOMATO - EXTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Investigate external structure in tomato using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-283 - TOMATO - INTERNAL STRUCTURE

Plant part	Fruit
Learning objective	Investigate internal structure in tomato using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-284 - TOMATO - COLOUR & TEXTURE

Plant part	Fruit
Learning objective	Investigate colour & texture in tomato using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-285 - TOMATO - MASS & SIZE

Plant part	Fruit
Learning objective	Investigate mass & size in tomato using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-286 - TOMATO - WATER/CHANGE OBSERVATION

Plant part	Fruit
Learning objective	Investigate water/change observation in tomato using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-287 - TOMATO - SORTING & QUALITY

Plant part	Fruit
Learning objective	Investigate sorting & quality in tomato using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-288 - TOMATO - POST-HARVEST CHANGE

Plant part	Fruit
Learning objective	Investigate post-harvest change in tomato using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-289 - TOMATO - INGREDIENT-TO-FOOD PATHWAY

Plant part	Fruit
Learning objective	Investigate ingredient-to-food pathway in tomato using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-290 - TOMATO - INVESTIGATION CHALLENGE

Plant part	Fruit
Learning objective	Investigate investigation challenge in tomato using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

INGREDIENT LAB - MAIZE | PLANT PART: SEED/GRAIN

Ten investigations develop progressively from ingredient identity and structure to quality, post-harvest change and an independent investigation challenge.

IL-291 - MAIZE - IDENTITY & PLANT PART

Plant part	Seed/Grain
Learning objective	Investigate identity & plant part in maize using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-292 - MAIZE - EXTERNAL STRUCTURE

Plant part	Seed/Grain
Learning objective	Investigate external structure in maize using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-293 - MAIZE - INTERNAL STRUCTURE

Plant part	Seed/Grain
Learning objective	Investigate internal structure in maize using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-294 - MAIZE - COLOUR & TEXTURE

Plant part	Seed/Grain
Learning objective	Investigate colour & texture in maize using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-295 - MAIZE - MASS & SIZE

Plant part	Seed/Grain
Learning objective	Investigate mass & size in maize using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-296 - MAIZE - WATER/CHANGE OBSERVATION

Plant part	Seed/Grain
Learning objective	Investigate water/change observation in maize using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-297 - MAIZE - SORTING & QUALITY

Plant part	Seed/Grain
Learning objective	Investigate sorting & quality in maize using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-298 - MAIZE - POST-HARVEST CHANGE

Plant part	Seed/Grain
Learning objective	Investigate post-harvest change in maize using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-299 - MAIZE - INGREDIENT-TO-FOOD PATHWAY

Plant part	Seed/Grain
Learning objective	Investigate ingredient-to-food pathway in maize using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?

IL-300 - MAIZE - INVESTIGATION CHALLENGE

Plant part	Seed/Grain
Learning objective	Investigate investigation challenge in maize using a fair sampling method, quantitative evidence and an evidence-based explanation.
Materials	Clean ingredient samples or images, labels, observation sheet/journal, ruler and balance where appropriate, clean trays/containers, handwashing supplies.
DIY / low-cost	Use locally available produce, paper labels, clean food-grade reused containers, handmade measuring grids and simple classroom balances where suitable.
Investigation method	Frame a question, select comparable samples, measure the relevant property, repeat observations, summarize the data and explain the pattern and one limitation.
Evidence to record	Question, sample IDs, data table, calculation/graph, conclusion and limitation.
Questions	Is the comparison fair? What variable are you measuring? What pattern is supported by the data?
Safety	No tasting is required. Use clean teacher-approved samples. Wash hands before and after handling. Teachers alone control knives, graters, heat, electrical equipment, chemicals and processing machinery. Do not conduct microbial culture or unsafe food-storage experiments.
AgriShine task	Connect the ingredient to where/how it could be grown, observed or studied in a school garden or food-literacy lesson.
Assessment link	Assess the relevant knowledge, practical competence, investigation/data skill or mission-based problem-solving dimension.

Lab reflection: What did the evidence show, what remains uncertain, and what would you investigate next?